Baby sunscreen, mineral UV filters (dirty): HMTc Standard

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CITE AS

Pendergrass K. "Baby sunscreen, mineral UV filters (dirty): HMTc Standard". Heavy Metal Certified, 2026. https://heavymetalcertified.-com/standards/baby-sunscreen-mineral. CC BY 4.0.

CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for baby sunscreen, mineral UV filters (dirty) in Infant and Child Personal Care (Ages 0-5), evaluated on finished cosmetic as placed on market (total content, ug/kg of product as sold). 5 of 11 panel analytes publish a numeric limit taken from the strictest applicable government maximum; the remainder are held under mandatory control or a substantiated not-material determination.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default) as applied to a child-scoped cosmetic category: every numeric cell is the strictest applicable sovereign finished-cosmetic maximum in the row's native total-content basis, or a disclosed read-across of a stricter sovereign tier from the same instrument where that instrument writes an adult-market loosening carve-out that HMTc declines to adopt for a product marketed to ages 0-5. No cell is occurrence-derived.

2 | Master limit table

All values in ppb (µg/kg), as_sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	2,000	government maximum level
Cadmium (Cd)	Tier 1	100	government maximum level

Total arsenic (tAs)	Tier 2	500	government maximum level
Inorganic arsenic (iAs)	Tier 1	≤ tAs	bounded by tAs (reflex speciation)
Mercury, total (tHg)	Tier 1	100	government maximum level
Methylmercury (MeHg)	Tier 1	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	Tier 2	10,000	government maximum level
Tin (Sn)	Tier 2	3,750,000	human-grade read-across (European Union)
Aluminium (Al)	Tier 2	560,000	human-grade read-across (European Union)
Total chromium (Cr)	Tier 2	50	human-grade read-across (AU)
Hexavalent chromium (Cr-VI)	Tier 2	5	human-grade read-across (European Union)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	2,000	government maximum level
Cadmium (Cd)	100	government maximum level
Total arsenic (tAs)	500	government maximum level
Inorganic arsenic (iAs)	≤ tAs	iAs carries no product-specific sovereign cell; it is a component of and bounded by its parent total tAs (iAs <= tAs), an enforceable ceiling via reflex speciation rather than an empty cell.
Mercury, total (tHg)	100	government maximum level

Methylmercury (MeHg)	≤ tHg	MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell.
Nickel (Ni)	10,000	government maximum level
Tin (Sn)	3,750,000	European Union — human-food analog
Aluminium (Al)	560,000	European Union — human-food analog
Total chromium (Cr)	50	AU — human-food analog
Hexavalent chromium (Cr-VI)	5	European Union — human-food analog

4 | Cite this standard

Pendergrass, K. "Baby sunscreen, mineral UV filters (dirty)." In Heavy Metal Tested & Certified (HMTc) Infant and Child Personal Care (Ages 0-5) standard, version 1.0 (2026-08-17). Institute of Contaminant Standards (ICS), 2026.

References

[1] AU — human-food maximum levels
Binds Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] European Union — human-food maximum levels
Binds Sn, Al, Cr-VI. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

Editorially reviewed in-house; no external peer review has been completed on this reference. Published as a preprint with its review state tracked on the article page. Released under CC BY 4.0.

COMPETING INTERESTS

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DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/-baby-sunscreen-mineral. Primary legal instruments and expert-body assessments are named with their public

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/-baby-sunscreen-mineral.

identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.